



Carbon UPI · Climate Compliance & Trust Infrastructure

GREEN IMPACT CERTIFICATE

Verified Emission Reduction — Formally Issued

ISSUED	HIGH CONFIDENCE 99/100	PUBLICLY VERIFIABLE	CBAM READY
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NET VERIFIED EMISSION REDUCTION	9.2761 tCO ₂ e
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Verification statement	9.2761 tCO ₂ e avoided — solar energy displacing grid electricity — Central Electricity Authority 0.82 kg CO ₂ /kWh — AMS-I.D — net of DG diesel Scope 1 (1525 litres)
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1. CERTIFICATE IDENTITY	
GIC ID	GP-GIC-2026-GJ-SOL-0442-Q3 FY2024-25
GIC Hash (SHA-256)	308201b5fc9fb9a4b5daafa02cb3b1be22f0aa8ef01dc92983dc4ac87e9ce645
Calculation Log Hash	c0b68c9a8e3f5c45b03685fb64da8f1defcd2a58144da61a9c6348694f6fa2a5
CIH Reference	GP-IND-2024-GJ-044821-SOL
Public Verification URL	https://verify.greenpe.in/gic/GP-GIC-2026-GJ-SOL-0442-Q3 FY2024-25
Ledger Anchor	Hedera Hashgraph — Transaction ID pending finalization
Certificate Status	ISSUED — Active and publicly verifiable
Issue Date	22-April-2026
Issuing Platform	GreenPe Technologies Pvt. Ltd., India

2. REGISTERED ENTITY	
Company Name	Surat Textiles Private Limited
GSTIN	24AADCS7412M1Z8 — VERIFIED
UDYAM / MSME Number	UDYAM-GJ-06-0044821 — VERIFIED
PAN	AADCS7412M — VERIFIED
Registered Address	Plot No. 44, GIDC Industrial Estate, Pandesara, Surat — 394221, Gujarat
Industry (NIC Code)	NIC-13111 — Preparation and Spinning of Textile Fibres
Contact (OTP Verified)	compliance@surattextiles.in +91-9876543210

3. PHYSICAL ASSET	
Asset Type	Rooftop Solar PV — Grid-tied

Export competitiveness	CBAM cost avoidance: ~€760 at €100/tonne — protects export margin on EU fabric orders
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ISSUER ATTESTATION

GreenPe Technologies Pvt. Ltd. certifies that this Green Impact Certificate was generated using the GreenPe Digital MRV Engine (Layer 3, Carbon UPI v1.0), with all source data verified through the CDIF standardisation layer (Layer 2). The emission reduction value is deterministic — produced by rule-based, version-controlled methodology logic — and is independently reproducible from the same inputs. The calculation log hash serves as an immutable audit trail.

This GIC is NOT a tradable carbon credit. It is a pre-issuance proof artifact that can be submitted to authorised registries (Verra, Gold Standard, BEE CCTS) for credit issuance, used as evidence for CBAM declarations, submitted as BRSR evidence, or consumed via API by banks for green finance KPI verification.

For GreenPe Technologies Pvt. Ltd.

Company Authorisation

[Authorised Signatory]

MRV Platform Lead — GreenPe

[Company Authorised Signatory]

Surat Textiles Private Limited

Date: 22-April-2026

GSTIN: 24AADCS7412M1Z8

INDEPENDENT VERIFICATION

Scan QR code or visit the URL below to independently verify this certificate. No login required

<https://verify.greenpe.in/gic/GP-GIC-2026-GJ-SOL-0442-Q3 FY2024-25>



Installed Capacity	50 kWp (kilowatt-peak)
Panel Configuration	91 × Waaree 550W Mono PERC panels
Inverter	Huawei SUN2000-50KTL-M3 (50 kW)
GPS Coordinates	21.1702°N, 72.8311°E — GPS verified at registration
Date of Commissioning	15-July-2024
Installer	Tata Power Solar Systems Ltd
Rooftop Area	280 sq. metres
IoT Device ID	HW-FUS-SRT-2024-0442 (Huawei FusionSolar API)
Device Fingerprint	Device-registered and fingerprint-verified — anti-cloning

4. MONITORING PERIOD	
Period Start	01-October-2024 (00:00:00 IST)
Period End	31-December-2024 (23:59:59 IST)
Duration	92 days — Q3 FY2024-25
Reporting Frequency	Daily IoT readings (15-minute intervals, daily batch)
Total Readings Expected	92 daily readings
Verified Readings	90 IoT-verified 2 estimated
Data Completeness	97.8% (90/92 verified)

5. METHODOLOGY & EMISSION FACTORS	
Methodology ID	AMS-I.D
Methodology Full Title	Grid Connected Renewable Electricity Generation — Small Scale
Methodology Version	Version 18.0 — UNFCCC CDM (2022)
Official Standard Body	UNFCCC CDM Verra VCS Compatible BEE CCTS Compatible
Grid Emission Factor (EFgrid)	0.82 kg CO2/kWh — Western Region
Grid EF Source	Central Electricity Authority, CO2 Baseline Database v19.0, FY2023-24
Diesel Emission Factor	2.68 kg CO2/litre (HSD) — IPCC 2006
Data Completeness Score	0.978 (97.8%)
Conservative Adj. Factor (CAF)	0.95 (5% discount — per AMS-I.D)
T&D Loss Factor	0.038
Additionality Test	PASSED — Common Practice Test per EB55 Annex II (penetration < 10%)

6. STEP-BY-STEP CALCULATION TRACE

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Step 1 — Solar energy generated	17540 kWh
Step 2 — Data completeness adj.	17154.12 kWh
Step 3 — Baseline emissions (raw)	14066.3784 kg CO ₂
Step 4 — Convert to tonnes	14.0664 tCO ₂ e
Step 5 — Apply CAF (5% discount)	13.3631 tCO ₂ e (× 0.95)
Step 6 — Scope 1 diesel (DG set)	1525 L × 2.68 kg/L = 4.087 tCO ₂ e
Step 7 — NET REDUCTION	9 . 2 7 6 1 t C O 2 e '

7. DATA QUALITY & CONFIDENCE

Identity Verification (CIH)	100/100 — GSTIN + UDYAM + GPS all bound
Data Completeness	97.8/100 — 90 of 92 IoT verified
Cross-Source Validation	96/100 — Inverter data vs DGVCL bill within ±2%
Methodology Adherence	100/100 — ZERO deviations
Anomaly Detection	98/100 — Zero anomalies
OVERALL CONFIDENCE SCORE	99 / 100 — HIGH CONFIDENCE

8. DOWNSTREAM USE CASES

EU CBAM Declaration	ENABLED — Emission intensity: 0.0225 tCO ₂ e/tonne fabric HSN 5208
Carbon Credit Registry Submission	ENABLED — Ready for Verra VCS CCTS India Gold Standard
SEBI BRSR Evidence	ENABLED — Principle 6, Indicator 14 verified
Green Loan / SLL KPI Proof	ENABLED — GIC API: GET /api/gic/GP-GIC-2026-GJ-SOL-0442-Q3 FY2024-25